

ICT Course: Introduction to Cryptography

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Session 6: Asymmetric Cryptography - RSA Cryptosystem and Diffie-Hellman Key Exchange

- 1 RSA Cryptosystem
 - Generating RSA public and private keypair
 - Encryption-Decryption
 - Proof of the correctness
 - Repeated squaring
 - Speed up RSA
 - Cryptanalysis
- 2 Diffie-Hellman Key exchange
- 3 Application of Public Key cryptography

Overview of RSA

- be made practical by Rivest, Shamir and Aldeman
- the most widely used asymmetric cryptosystem
- Applications: Transport of (symmetric) keys and Digital Signature

RSA Key Generation

- Choose 2 large prime numbers p, q : $N = p * q$
- Choose e relative prime to $(p - 1)(q - 1)$
- Find d :

$$ed = 1 \bmod (p - 1)(q - 1) \iff e^{-1} \bmod (p - 1)(q - 1) = d$$

- RSA key pair consists of:
 - Public key: (N, e)
 - Private key: dwhere N : modulus, e : encryption exponent, d : decryption exponent
- N has 1024 bits or 2048 bits or larger

Encryption - Decryption

- Encryption: $C \equiv M^e \pmod{N}$
- Decryption: $M \equiv C^d \pmod{N}$

Proof of correctness

Euler's phi function reminds:

- $\phi(m)$ is the number of positive integers less than m that are relatively prime to m
- For any prime number p : $\phi(p) = (p - 1)$
- If p, q are prime: $\phi(p * q) = (p - 1)(q - 1)$

Euler's Theorem

If x is relative prime to n then $x^{\phi(n)} \equiv 1 \text{ mod } n$

Proof of correctness

- Assume that M is relatively prime to N , proof the correctness of RSA?
- If M is not prime to N , proof of the correctness of RSA?

Textbook RSA example

- $p = 11, q = 3$, choose $e = 3$
- Key pairs?
- Describe the encryption and decryption using RSA if Bob want to send a plaintext $M = 15$ to Alice?

- Example: $(N, e) = (33, 23)$, $M = 5$, Calculate C
- Problem?
- Repeated Squaring method

- Using same exponent e for all users and different p, q are chosen for each key pair
- Common used encryption exponent:
 - $e = 3$: requires $M > N^{1/3}$ to avoid **cube root attack**
 - $e = 2^{16} + 1$

Cryptanalysis

- Protocol attack
- Mathematical attack
- Side-channel attack

Diffie-Hellman Key exchange - Overview¹

- Proposed in 1976 by Whitfield Diffie and Martin Hellman
- Widely used, e.g. in Secure Shell (SSH), Transport Layer security (TLS), Internet Protocol Security (IPSec)
- Used to establish a shared key, not usually for encryption

¹Understanding Cryptography by Christof Paar and Jan Pelzl

Diffie-Hellman Key Exchange

Discrete Logarithm problem:

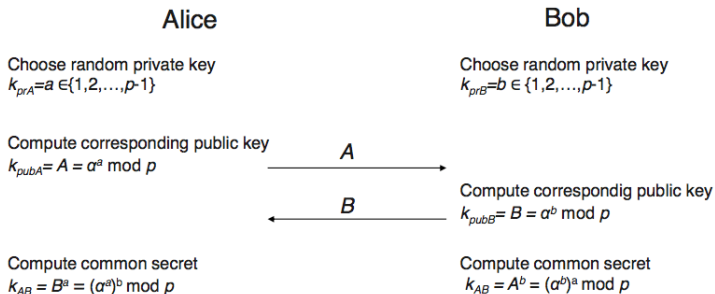
Given integers $g, p, g^k \bmod p$, find k

$\implies$ Very difficult to solve

Diffie-Hellman Key Exchange setup:

- Choose a large prime p and a (integer) generator g
- $\forall x \in \{1, 2, \dots, p-1\}, \exists n : x \equiv g^n \pmod{p}$
- g, p are public

Diffie-Hellman Key Exchange(DHKE)



We can now use the joint key k_{AB}
for encryption, e.g., with AES

$$y = AES_{k_{AB}}(x) \quad \xrightarrow{y} \quad x = AES_{k_{AB}}^{-1}(y)$$

DHKE - Man-in-Middle Attack

- Confidentiality:
 - use key pairs to encrypt data
 - hybrid cryptosystem
- Integrity: Digital signature